

Norbert O. Schedler Honors College Mentor-Mentee Agreement

Mentor-Mentee Agreement Instructions

Please complete this Mentor-Mentee agreement after first meeting with your mentor and determining the following:

Meeting schedule: frequency, days, times, and location

Semester goals: what will you accomplish this semester?

Reading list: work together with your mentor to curate a list of works that you will read this semester to move your project forward.

You do not have to complete the form in one sitting -- you can save your progress and return to it later. Once you complete the form, you will sign and submit it. The form will then be routed to your mentor for their signature. They do have the option to return the form to you for corrections, so please be as specific as possible when completing the form. Once your mentor signs the form, you will get a confirmation email notifying you of their signature. After your mentor signs, the form will be routed to the Honors Undergraduate Research Coordinator, Tina Huddleson. Tina will review the form and also has the ability to return it to you for corrections.

The completed forms are due by Friday, September 12. For students enrolled in Oxford Tutorial this semester, your deadline is Friday, October 10. Please reach out to Tina if you have any questions, thuddleson@uca.edu.

By initialing below, I confirm that I have read the instructions for completing the Honors Mentor-Mentee Agreement.

I confirm that I have read the Mentor-Mentee Agreement Instructions

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Mentee First Name: *

Last Name: *

StudentID *

Mentor First Name: *

Mentor Last Name: *

Mentor Email (make sure email is entered correctly) *

Project Title *

Expected Capstone Semester *

Provide a brief project description. Feel free to use language from your proposal abstract.

Project Description

The recent prominence of Large Language Models has emphasized the greater needs for improvements in the field of Natural Language Inference. The abilities of various Large Language Models on understanding entailment need to be expanded from the current benchmarks in order to better reflect a larger array of possible use cases of human language. As such, this capstone project seeks to answer the three questions research questions: (1) What current cases of entailment is the BERT Model left vulnerable to misinterpretation because the MultiNLI benchmark doesn't cover it, (2) How can these voids be exposed via a modification of the MultiNLI benchmark into a novel data-set featuring ungrammaticality, and (3), how will the BERT Model, when tested on the original MultiNLI, perform on the modified MultiNLI, and what could that say about BERT Model's NLI abilities?

General Expectations Mentors and mentees will have regular meetings in which the mentee updates the mentor on their progress, reading, ideas, and questions. The mentor will question and clarify the work in order to move the project forward so that by the time the student enrolls in the Honors Capstone course, the project/research component is complete and the student is ready to reflect, analyze data, and write up the project.

Meeting Schedule (days, times, locations) *Every Wednesday at 10:30am to Noon in MCS 331 and any other times scheduled as necessary

What will you accomplish this semester? Be specific with tasks and deadlines. *The project goals for Fall 2025 are:
— Have CLS tokens from test data set pickled via python pickle package by 9/13/2025 to test if pickling and reopening objects in a new file will work properly
— Write a concatenation and evaluation file in python that will work with the test data by 9/20/2025
— Pickle the CLS tokens from the real data set and transfer them to the concatenation and evaluation file by 9/27/2025
— Work with K-Cross-Fold values in the concatenate and evaluation function to get presentable results about the performance of the BERT model on the modified data set vs the original data set by 10/4/2025
— Write up initial report of the data and save it 10/10/2025
— Repeat this experiment with other modifications (removal of words beside the prepositions) if time is available 10/17/2025
— Continue to write up reports and document project 10/24/2025
— Set up the Github and have presentable format of the project summary that could be submitted as a sample of written for graduate school applications 10/31/2025
— The month of November exists as a buffer in case there are delays, but I HAVE TO HAVE THIS DONE before the end of October!

Reading Goals The student will read works of their own finding and works suggested by the mentor. List specific works below.

ReadingList *The mentee will read works suggested by the mentor and other works that the student may find relevant to the study. This semester, specific works include:

Russell, J. Stuart & Norvig, Peter. "Artificial Intelligence: A Modern Approach" Pearson. 4th ed. 2021.

"Understanding Python Pickling with example" GeeksforGeeks. 14 Aug. 2024. <https://www.geeksforgeeks.org/python/understanding-python-pickling-example/>

Verma, Dhruv. et al. "Evaluating Paraphrastic Robustness in Textual Entailment Models" ArXiv. 29 Jun. 2023. <https://arxiv.org/abs/2306.16722>

Liu, Alisa. et al. "WaNLI: Worker and AI Collaboration for Natural Language Inference Dataset Creation." ArXiv. <https://arxiv.org/abs/2201.05955>

Courneau, David, et al. "LogisticRegression." Scikit Learn. 2007. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

And may also include other readings as assigned.

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Kyle Urban 09/10/2025
Mentee Signature Date

Mentor Signature Date

Capstone Mentor

Honors Undergraduate Research Coordinator